CS2108
AY24/25 SEM 2
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Sinusoids

- Sinusoids are periodic functions that can be represented as a sum of sine and cosine functions. The general form of a sinusoid is:

$$y(t) = A\sin(\omega t + \phi) + B\cos(\omega t + \phi) \tag{1}$$

where:
  - A is the amplitude
  - ω is the angular frequency
  - ϕ is the phase shift
  - B is the vertical shift
- The angular frequency ω is related to the frequency f and the period T by:

$$\omega = 2\pi f = 2\pi \frac{1}{T} \tag{2}$$

- Hence, the sinusoid can also be expressed in terms of frequency:

$$y(t) = A\sin(2\pi f t + \phi) + B\cos(2\pi f t + \phi) \tag{3}$$

Discrete Time Sinusoids

- Discrete time sinusoids are sampled versions of continuous time sinusoids. The general form is:

$$x[n] = \sum_{k=0}^{K-1} a_k \sin(2\pi f_k t_n) + b_k \cos(2\pi f_k t_n) \tag{4}$$

where:
  - n is the discrete time index
  - k is the frequency index
  - t_n is the discrete time variable $n\Delta t$
  - f_k is the frequency of the k-th component $k\Delta f$
  - Δt is the sampling period
  - Δf is the frequency resolution
  - K is the total number of frequencies

  The coefficients a_k and b_k define the amplitude and phase of each frequency component.
- The discrete time sinusoid can also be expressed in different forms:

$$\begin{aligned} x[n] &= \sum_{k=0}^{K-1} a_k \sin(2\pi f_k t_n) + b_k \cos(2\pi f_k t_n) \\ &= \sum_{k=0}^{K-1} c_k \sin(2\pi f_k t_n + \phi_k) \\ &= \sum_{k=0}^{K-1} d_k \cos(2\pi f_k t_n + \theta_k) \end{aligned} \tag{5}$$

where:
  - Second line is the same sinusoid a summation of shifted sine functions
  - Third line is the same sinusoid as a summation of shifted cosine functions
  - c_k and d_k are the coefficients for the sine and cosine forms
  - ϕ_k and θ_k are the phase shifts for the sine and cosine forms
  - The relationship between the coefficients is given by:

$$\begin{aligned} c_k &= \sqrt{a_k^2 + b_k^2} \\ d_k &= \sqrt{a_k^2 + b_k^2} \\ \phi_k &= \tan^{-1}\left(\frac{b_k}{a_k}\right) \\ \theta_k &= \tan^{-1}\left(\frac{a_k}{b_k}\right) \end{aligned} \tag{6}$$

  - If a_k = b_k, the sinusoid is a combination of sine and cosine functions with equal amplitude and phase shift. Phase shift is 45 degrees.
  - If a_k = -b_k, the sinusoid is a combination of sine and cosine functions with equal amplitude and phase shift. Phase shift is 135 degrees.
  - If a_k = 0, phase shift is 90 degrees
  - If b_k = 0, phase shift is 0 degrees

Periodicity

- Fundamental period is the smallest period of a sinusoid. The fundamental period is the time it takes for the sinusoid to complete one full cycle.
- The fundamental frequency is the reciprocal of the fundamental period:

$$f_0 = \frac{1}{T_0} \tag{7}$$

- The fundamental frequency is the lowest frequency of a sinusoid. It is the frequency at which the sinusoid oscillates.
- Sinusoids are periodic functions, meaning they repeat their values at regular intervals.
- If we add sinoiuds with different frequencies, the resulting function is not periodic unless the frequencies are rationally related.
- Suppose we have a sinusoid with the equation:

$$y(t) = a_1 \sin(\omega_1 t + \phi_1) + a_2 \cos(\omega_2 t + \phi_2) \tag{8}$$

- To determine if the sinusoid is periodic, we need to check if the ratio of the **frequencies** is rational:

$$\frac{\omega_1}{\omega_2} = \frac{p}{q} \tag{9}$$

where p and q are integers.
- To determine the **fundamental period** of the sinusoid, we need to find the least common multiple (LCM) of the periods of the individual sinusoids:

$$T = \text{lcm}(T_1, T_2) \tag{10}$$

where T_1 and T_2 are the periods of the individual sinusoids. Calculate the period of each sinusoid using:

$$T_i = \frac{2\pi}{\omega_i} \tag{11}$$

- Hint: Let ratio of periods = $\frac{T_1}{T_2} = \frac{q}{p}$, then $T = pT_1 = qT_2$
- Suppose we have a sinusoid with the equation:

$$y(t) = a_1 \sin(2\pi f_1 t + \phi_1) + a_2 \cos(2\pi f_2 t + \phi_2) \tag{12}$$

where f_1 and f_2 are integers
- To determine the **fundamental frequency** of the sinusoids, we need to find the greatest common divisor (GCD) of the frequencies of the individual sinusoids:

$$f = \text{gcd}(f_1, f_2) \tag{13}$$

- Then fundamental period is: $T = \frac{1}{f}$
- The above calculatiosn can be done with sinusoids of more than 2 frequencies as well.

Fourier Series

Recap

- **Orthogonal Basis Set** a set of functions that are orthogonal to each other. This means that the inner product of any two functions in the set is zero.
- Inner product of two functions $f(t)$ and $g(t)$ is defined as:

$$\langle f(t), g(t) \rangle = \int_{-\infty}^{\infty} f(t)g(t)dt \tag{14}$$

- Inner Product (continuous) = Dot Product (discrete)
- **Sine and cosine functions are orthogonal to each other**, no matter their frequencies. Inner/Dot product = **zero**.

  Let $f_m = mf_0$ and $f_n = nf_0$ where m and n are integers

$$\int_0^T \sin(2\pi f_m t)\cos(2\pi f_n t)\,dt = \frac{1}{2}\left[\frac{\cos(2\pi(f_m - f_n)t)}{-2\pi(f_m - f_n)} - \frac{\cos(2\pi(f_m + f_n)t)}{2\pi(f_m + f_n)}\right]_0^T = \begin{cases} 0 & if\ m \neq n \\ 0 & if\ m == n \end{cases}$$

- **Sine waves of different frequencies** are orthogonal to each other.Inner/Dot product = **zero**.

  Let $f_m = mf_0$ and $f_n = nf_0$ where m and n are integers

$$\int_0^T \sin(2\pi f_m t)\sin(2\pi f_n t)\,dt = \frac{1}{2}\left[\frac{\sin(2\pi(f_m - f_n)t)}{2\pi(f_m - f_n)} - \frac{\sin(2\pi(f_m + f_n)t)}{2\pi(f_m + f_n)}\right]_0^T = \begin{cases} 0 & if\ m \neq n \\ \frac{T}{2} & if\ m == n \end{cases}$$

- Similarly, **Cosine waves of different frequencies** are orthogonal to each other. Inner/Dot product = **zero**.

  Let $f_m = mf_0$ and $f_n = nf_0$ where m and n are integers

$$\int_0^T \cos(2\pi f_m t)\cos(2\pi f_n t)\,dt = \frac{1}{2}\left[\frac{\sin(2\pi(f_m - f_n)t)}{2\pi(f_m - f_n)} + \frac{\sin(2\pi(f_m + f_n)t)}{2\pi(f_m + f_n)}\right]_0^T = \begin{cases} 0 & if\ m \neq n \\ \frac{T}{2} & if\ m == n \end{cases}$$

- For sine and cosine with non-zero phase shifts:

$$\begin{aligned} \sin(2\pi f t + \phi) &= sin(2\pi f t)\cos(\phi) + \cos(2\pi f t)\sin(\phi) \\ \cos(2\pi f t + \phi) &= cos(2\pi f t)\cos(\phi) - \sin(2\pi f t)\sin(\phi) \end{aligned} \tag{15}$$

- In other words, with a phase shift, a sine or cosine function that is phase shifted can be expressed as a weighted sum of sine and cosine functions.
- So in general, the sum of sinusoidal signals, even with non-zero (but constant) phase, can be written as a linear combination of sine and cosine with zero phase.

$$\textstyle\sum_{k=0}^{K-1} A_k \sin(2\pi f_k t + \phi_k) = \sum_{k=0}^{K-1} A_k \left[\sin(2\pi f_k t)\cos(\phi_k) + \cos(2\pi f_k t)\sin(\phi_k)\right]$$

$$\textstyle\sum_{k=0}^{K-1} B_k \cos(2\pi f_k t + \phi_k) = \sum_{k=0}^{K-1} A_k \left[\cos(2\pi f_k t)\cos(\phi_k) - \sin(2\pi f_k t)\sin(\phi_k)\right]$$

Fourier Series

- Fourier series is a way to represent a periodic signal as a sum of sine and cosine functions. It is used to analyze periodic signals in the frequency domain.

$$y(t) = \sum_{m=0}^{M-1} (a_m \sin(2\pi f_m t) + b_m \cos(2\pi f_m t)) \tag{16}$$

- How do we determine all of the a_m and b_m:

  Recall that we can represent any periodic signal by a linear combination of shifted sines, i.e.

$$\begin{aligned} any.periodic.signal &= \sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m) \\ &= \sum_{m=0}^{M-1} A_m \left(\sin(2\pi f_m t)\cos(\phi_m) + \cos(2\pi f_m t)\sin(\phi_m)\right) \\ &= \sum_{m=0}^{M-1} \left[A_m\cos(\phi_m)\sin(2\pi f_m t) + A_m\sin(\phi_m)\cos(2\pi f_m t)\,(\quad)\right. \\ &= \sum_{m=0}^{M-1} \left[\underbrace{A_m\cos(\phi_m)}_{a_m}\sin(2\pi f_m t) + \underbrace{A_m\sin(\phi_m)}_{b_m}\cos(2\pi f_m t)\right] \end{aligned}$$

$$\begin{aligned} a_m &= A_m \cos(\phi_m) \\ b_m &= A_m \sin(\phi_m) \end{aligned} \tag{17}$$

- The Fourier series coefficients a_m and b_m are calculated using the following formulas:

$$\begin{aligned} a_m &= \frac{2}{T}\int_0^T y(t)\sin(2\pi f_m t)dt \\ b_m &= \frac{2}{T}\int_0^T y(t)\cos(2\pi f_m t)dt \end{aligned} \tag{18}$$

Recall the following integration results [Dot Product]

Mixed Signal

$$\int_0^T \left[\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)\right]\sin(2\pi f_k t)\,dt = \frac{T}{2}A_k\cos(\phi_k)$$

$$\int_0^T \left[\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)\right]\cos(2\pi f_k t)\,dt = \frac{T}{2}A_k\sin(\phi_k)$$

$$\int_0^T \left[\sum_{m=0}^{M-1} B_m \cos(2\pi f_m t + \phi_m)\right]\sin(2\pi f_k t)\,dt = \frac{-T}{2}A_k\sin(\phi_k)$$

$$\int_0^T \left[\sum_{m=0}^{M-1} B_m \cos(2\pi f_m t + \phi_m)\right]\cos(2\pi f_k t)\,dt = \frac{T}{2}A_k\cos(\phi_k)$$

We can therefore use $\sin(2\pi f_k t)$ and $\cos(2\pi f_k t)$ to figure out what are all the a_m and b_m in the signal.

$$\int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \sin(2\pi f_k t)\, dt = \frac{T}{2} A_k \cos(\phi_k) = \frac{T}{2} a_k$$

$$\int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \cos(2\pi f_k t)\, dt = \frac{T}{2} A_k \sin(\phi_k) = \frac{T}{2} b_k$$

- The ratio of b_k to a_k gives the phase shift of the sinusoid:

$$\frac{b_k}{a_k} = \frac{A_k \sin(\phi_k)}{A_k \cos(\phi_k)} = tan(\phi_k)$$
$$\phi_k = \tan^{-1}\left(\frac{b_k}{a_k}\right) \tag{19}$$

- We can use complex numbers to represent the Fourier series:

$$X(f_k) = \int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \cos(2\pi f_k t)\, dt$$
$$-j \int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m)] \sin(2\pi f_k t)\, dt$$
$$= \frac{T}{2} A_k \sin(\phi_k) \text{ - j} \frac{T}{2} A_k \cos(\phi_k)$$
$$= \frac{T}{2} (\, b_k \text{ - j } a_k\,)$$

- **Magnitude** = $|X_k| = \frac{T}{2}\sqrt{a_k^2 + b_k^2}$
- **Phase** = $\tan^{-1}\left(\frac{b_k}{a_k}\right)$
- We can also express the Fourier series in terms of **complex exponentials**, recall Eulers formula: $e^{j\theta} = \cos(\theta) + j\sin(\theta)$, then:

$$X(f_k) = \int_0^T [\sum_{m=0}^{M-1} A_m \sin(2\pi f_m t + \phi_m] e^{-j2\pi f_k t} dt \tag{20}$$

- **Fourier series formula** (important):

$$X(f_k) = \int_0^T x(t) e^{-j2\pi f_k t} dt \tag{21}$$

Fourier Transform

- Fourier transform is a way to represent a non-periodic signal as a sum of sine and cosine functions. It is used to analyze non-periodic signals in the frequency domain.
- **Intuition:** For non-periodic signals, we can think that its *period is infinite*. Then the whole signal can be imagined as periodic with infinite period.
- Fourier series comparison:

$$X(f) = \int_0^T x(t) e^{-j2\pi f t} dt \tag{22}$$

- Fourier transform is defined as:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt \tag{23}$$

- The inverse Fourier transform is defined as:

$$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi f t} df \tag{24}$$

- The Fourier transform can be used to analyze the frequency content of a signal. It can be used to determine the amplitude and phase of each frequency component in the signal.
- Fourier transform equation using radian frequency ω:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \tag{25}$$

- The inverse Fourier transform using radian frequency ω:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \tag{26}$$

- The Fourier transform can also be used to filter signals. By multiplying the Fourier transform of a signal by a filter function, we can remove or enhance certain frequency components in the signal.

Discrete Fourier Transform (DFT)

- **DFT** Formula:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi k \frac{n}{N}} \tag{27}$$

- **Inverse DFT** Formula:

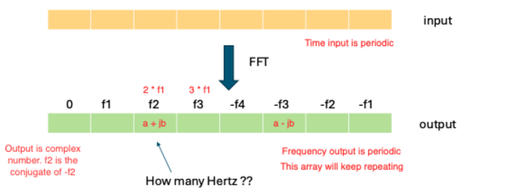
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi k \frac{n}{N}} \tag{28}$$

  where:
  - N is the number of samples
  - k is the frequency index (discretised frequencies)
  - n is the time index
  - $x[n]$ is the input signal
  - $X[k]$ is the DFT of the input signal
- FFT (Fast Fourier Transform) is an efficient algorithm to compute the DFT. It reduces the computational complexity from $O(N^2)$ to $O(N \log N)$.

Frequency Bins

- Frequency resolution: Smallest frequency difference that can be resolved by the DFT. It is determined by the sampling frequency and the number of samples.
- The DFT maps the input signal to a set of frequency bins. Each bin corresponds to a specific frequency range.

When you input a vector of, say, 8 numbers into an FFT function, and you get an output vector of length 8, how do you interpret the output vector?



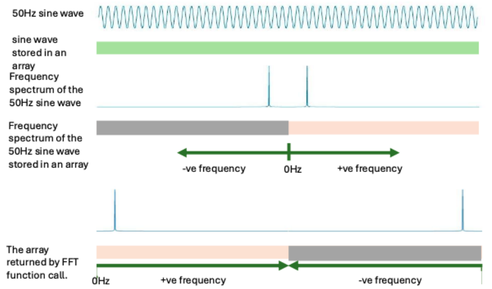
- The frequency resolution of the DFT is given by:

$$\Delta f = \frac{f_s}{N} \tag{29}$$

  where:
  - f_s is the sampling frequency
  - N is the number of samples
- The frequency bins are given by:

$$f_k = k\Delta f \tag{30}$$

- The DC component (k=0) corresponds to the average value of the signal. It is a constant in the sinusoid equation. The Nyquist frequency (k=N/2) corresponds to the highest frequency that can be represented by the DFT.
- We can imagine the DC component as vertical offset of the signal from the x-axis.
- **Visualisation**:



- Derivation of the frequency bins:

$$H(f) = \int_{t=-\infty}^{\infty} h(t)\, e^{-j2\pi ft}\, dt$$

$$H(f) \approx \sum_{n=0}^{N-1} h[n]\, e^{-j2\pi f n \Delta t}\, \Delta t$$

$$H(f) \approx \Delta t \sum_{n=0}^{N-1} h[n]\, e^{-j2\pi f n \Delta t}$$

$$H(f) \approx \frac{1}{f_s} \sum_{n=0}^{N-1} h[n]\, e^{-j2\pi \frac{f}{f_s} n}$$

Let $f = k\,\Delta f$, i.e. we also discretize the frequency axis using a frequency step Δf

$$H[k] \approx \frac{1}{f_s} \sum_{n=0}^{N-1} h[n]\, e^{-j2\pi \frac{k\Delta f}{f_s} n}$$

Recall that $H[k] \approx \sum_{n=0}^{N-1} h[n]\, e^{-j2\pi \frac{k}{N} n}$

Ignoring the scale factor, we have

$$2\pi \frac{k\Delta f}{f_s} n = 2\pi \frac{k}{N} n$$

Therefore,

$$\frac{\Delta f}{f_s} = \frac{1}{N}$$

- **Note**: Sometimes, we pad the input signal with zeros to increase the number of samples. Padding zeros to the input decreases the frequency bin size, but does not increase the frequency resolution.
- **Note**: N = 2^n for FFT, where n is an integer. This is because the FFT algorithm is based on the divide-and-conquer approach, which works best when the number of samples is a power of 2.

Properties of DFT

- **Intuition**: When you discretise one domain, you also discretise the other domain. The *other domain becomes periodic.*

- **Linearity**: $\mathcal{F}\{ax[n] + by[n]\} = aX[k] + bY[k]$
- **Time Shifting**: $\mathcal{F}\{x[n - n_0]\} = X[k]e^{-j2\pi k \frac{n_0}{N}}$
- If you shift a signal $x[n]$ by m samples in time, the Fourier transform $X[k]$ remains the same in magnitude, but is multiplied by a complex exponential.
- This complex exponential introduces a linear phase shift in the frequency domain. This exponential has a magnitude of 1, so the magnitude spectrum stays unchanged.
- **Frequency Shifting**: $\mathcal{F}\{x[n]e^{j2\pi m \frac{n}{N}}\} = X[k - m]$
- If you multiply $x[n]$ by a complex exponential in the time domain, it causes a circular shift in the frequency domain. Shifting the frequency response by m bins. Both magnitude and phase are affected.
- **Time Reversal**: $\mathcal{F}\{x[-n]\} = X^*[k]$
- **Conjugate Symmetry**: $X[k] = X^*[N - k]$
- Here $X^*[k]$ is the complex conjugate of $X[k]$. (i.e If $X[k] = a + jb$, then $X^*[k] = a - jb$)
- It is important to note that the $|X[k]| = |X^*[k]|$

- Conversion **from continuous to discrete**:

Let $t = n\,.\,\Delta t = n\,\frac{T}{N}$ where N is the total number of points

$$-\infty < t < \infty \quad \rightarrow \quad -\infty < n < \infty$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\; e^{-j\omega t}\, dt \quad \rightarrow \quad X(\omega) = \sum_{n=-\infty}^{\infty} x[n]\; e^{-j\omega n.\Delta t}\, \Delta t$$
$$= \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n]\; e^{-j\omega \frac{nT}{N}}$$

- **Note**: $\Delta t = \frac{T}{N}$

Let $\omega = k\,\omega_0 = k\; 2\pi f_0 = k\; 2\pi \frac{1}{T}$

$$X[k] = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n]\; e^{-jk2\pi \frac{1}{T}\frac{nT}{N}}$$

$$X[k] = \frac{T}{N} \sum_{n=-\infty}^{\infty} x[n]\; e^{-j2\pi k \frac{n}{N}}$$

- DFT output is **periodic with period N** (repeats every N samples). $X[k] = X[k + N]$

$$X[k] = \frac{T}{N}\sum_{n=-\infty}^{\infty} x[n]\ e^{-j2\pi k\frac{n}{N}}$$

$$X[k+N] = \frac{T}{N}\sum_{n=-\infty}^{\infty} x[n]\ e^{-j2\pi (k+N)\frac{n}{N}}$$

$$= \frac{T}{N}\sum_{n=-\infty}^{\infty} x[n]\ e^{-j2\pi k\frac{n}{N} - j2\pi n}$$

e^{-j*2*pi*n}
= cos(2*pi*n) - j sin(2*pi*n)
= 1

$$= \frac{T}{N}\sum_{n=-\infty}^{\infty} x[n]\ e^{-j2\pi k\frac{n}{N}}$$

$$= X[k] \longrightarrow X[k] \text{ is periodic with period N}$$

- Similarly, the input signal is periodic with period N.
  $x[n] = x[n+N]$

$$x[n] = \frac{1}{N}\sum_{k=-\infty}^{\infty} X[k]e^{j2\pi k\frac{n}{N}} \quad (31)$$

- **Summary:**

continuous time, continuous frequency

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\ e^{-j\omega t}\, dt$$

CTFT

Continuous Time Fourier Transform

discrete time, continuous frequency (i.e. ω is continuous)

Used in Sampling

$$X(\omega) = \frac{T}{N}\sum_{n=-\infty}^{\infty} x[n]\ e^{-j\omega\frac{nT}{N}}$$

DTFT

Discrete Time Fourier Transform

Time axis is discretized but frequency axis is not

discrete time, discrete frequency (i.e. k is discrete)

NOTE: the N here is not the period in seconds. The N here is an integer representing the total number of sampled points within one period.

$$X[k] = \sum_{n=0}^{N-1} x[n]\ e^{-j2\pi k\frac{n}{N}}$$

A/Prof NG Teck Khim

DFT (FFT)

Discrete Fourier Transform (DFT)
(Fast Fourier Transform is a fast implementation of DFT)

continuous time, continuous frequency

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\ e^{j\omega t}\, d\omega$$

Inverse CTFT

discrete time, continuous frequency (i.e. k is continuous)

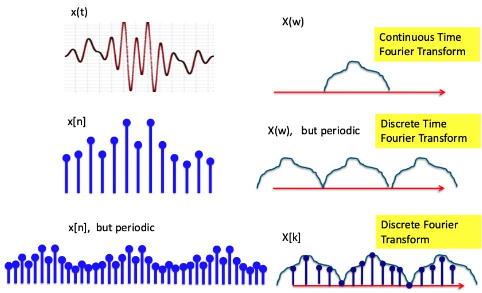
$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(\omega)\ e^{j\omega\frac{nT}{N}}\, d\omega$$

Inverse DTFT

discrete time, discrete frequency (i.e. k is discrete)

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\ e^{j2\pi k\frac{n}{N}}$$

Inverse DFT(iFFT)



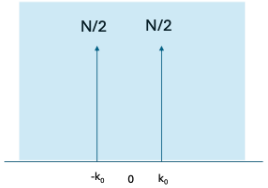
FFT is fast version of Discrete Fourier Transform

Useful Fourier Transform Results

- Fourier transform of an **impulse** in the time domain will result in a constant function in the frequency domain.
  [1, 0, 0, 0] $\rightarrow$ [1, 1, 1, 1]
- Fourier transform of a **constant function** in the time domain will result in an impulse in the frequency domain.
  [1, 1, 1, 1] $\rightarrow$ [4, 0, 0, 0]

Discrete Fourier Transform Pairs

$$\cos\left(\frac{2\pi}{N}k_o n\right) \underset{\text{iFFT}}{\overset{\text{FFT}}{\rightleftarrows}} \frac{N}{2}\left(\delta[k-k_o] + \delta[k+k_o]\right)$$



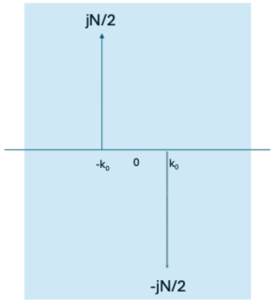
Only real component

Amplitude of N/2

Fundamental Frequency, will have a spike

Discrete Fourier Transform Pairs

$$\sin\left(\frac{2\pi}{N}k_o n\right) \underset{\text{iFFT}}{\overset{\text{FFT}}{\rightleftarrows}} \frac{-jN}{2}\delta[k-k_o] + \frac{jN}{2}\delta[k+k_o]$$

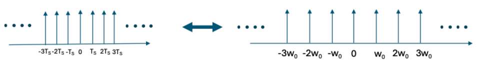


Only imaginary component

Continuous Time Fourier Transform for an Impulse Train

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow P(\omega) = \sum_{k=-\infty}^{\infty} \omega_0\, \delta(\omega - k\omega_0)$$

where $\omega_0 = \frac{2\pi}{T_s}$



Continuous Time Fourier Transform for a Rectangular Function

$$rect(t) = \begin{cases} 1 & -0.5 \le t \le 0.5 \\ 0 & otherwise \end{cases}$$

L'Hopital's Rule

$$x(t) = rect(t) \longleftrightarrow X(\omega) = \frac{2\sin(\frac{\omega}{2})}{\omega} = sinc\left(\frac{\omega}{2\pi}\right)$$